

WORK EXPERIENCE

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- CrossRefHVAC** · crossrefhvac.com 2026–Present
- *Founder & Software Engineer, Barrett Technologies LLC*
Python, PostgreSQL/Supabase, Parquet, Next.js, TypeScript, pdfplumber, pytest, Vercel
HVAC/R equipment data platform, launching publicly as SpecTrace. Turns manufacturer documentation and product APIs into cross-referenced, queryable data with per-field provenance, so machine extraction stays auditable at scale; coverage architecture sized for an 8M-fact, multi-brand catalog.
 - Built a test-driven Python pipeline (378 tests) that extracts structured product data from dense manufacturer PDFs: spec tables, nomenclature systems, and model families across 100+ change-tracked source documents.
 - Mapped an undocumented manufacturer product API via agent-assisted discovery: 684 configurations across three brand catalogs and 27K+ SKUs in the first vendor rollout, with convergence probes to prove full coverage.
 - Built ground-truth verification: pipeline output is checked against an independent product index, human rulings persist in an adjudication ledger, and unverified data cannot ship.
 - Designed an OLAP fact-table schema in PostgreSQL, validated against human-curated ground truth and exported as versioned columnar Parquet snapshots.
 - Built a human-in-the-loop review UI that overlays extraction results on the source PDF; each ruling is recorded once and reapplied automatically.
 - Serve the dataset through a REST API and an MCP server; every value carries provenance to its source document and page, so AI agents can query equipment data with citations.
- No Plastic, No Problem.** · noplasticnoproblem.com 2023–Present
- *Founder*
Next.js, React, TypeScript, Tailwind, Sanity CMS, n8n, Claude API, PostHog, Sentry, Supabase, Vercel
Production affiliate content platform serving 10,000+ monthly users with 50+ brand partnerships.
 - Built and operate a production affiliate platform serving 10,000+ monthly users, with a custom TypeScript/Sanity pipeline managing 200+ content entities and a bidirectional Notion integration powering the full editorial workflow.
 - Built an end-to-end content automation pipeline using n8n, DataForSEO, and the Claude API, cutting SEO brief generation from hours to automated output and supporting a 3x increase in content throughput.
 - Reduced CDN request volume by 85% and eliminated compute cost spikes by migrating to webhook-driven ISR revalidation, with automated search index regeneration triggered on every content publish.
 - Built a custom MCP server connecting Sanity and Notion as a live knowledge graph, giving AI workflows real-time access to article inventory and brand relationships for fully grounded content generation.
- **Haven Technologies** New York, NY
Fullstack Software Engineer June 2022 - August 2023
 - Redesigned and rebuilt an insurance policy viewer into a scalable SaaS product using Angular, PostgreSQL, Node.js, and Express, with test coverage via Mocha, Chai, and Sinon.
 - Implemented NgRx for state management and centralized API call patterns, improving consistency and debuggability across the frontend codebase.
- **Allvue Systems** New York, NY
Application Engineer, Operations July 2020 - May 2022
 - Resolved TSQL, XML, and XSLT errors in custom portfolio management software, improving reliability for B2B clients.
 - Wrote scripts to fix trade data and payment processing inconsistencies, ensuring accuracy in compliance assessments.

EDUCATION

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- **Stevens Institute of Technology** Hoboken, NJ
Bachelor of Science in Computer Science September 2016 - May 2020

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Python, SQL, HTML/CSS
Frameworks & Libs.	React, Next.js, Angular, Redux, NgRx, RxJS, Express, Node.js, Tailwind CSS
Tools	Git, GitHub, Vercel, Sanity, n8n, Sentry, PostHog, pytest, Vitest, pdfplumber, Docker, AWS, Jira
Data	PostgreSQL, Supabase, Parquet, MongoDB, Oracle, MySQL